



For Performance Measurement

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL

General Certificate of Education Advanced Level

PHYSICS

PAPER 1 Multiple Choice

6032/1

1 hour

JUNE 2026 SESSION

Additional materials:

Multiple Choice answer sheet

Soft clean eraser

Soft pencil (type B or HB is recommended)

Electronic calculator 86DV

INSTRUCTIONS TO CANDIDATES

Answer **all** questions. For each question, there are **four** possible answers, **A, B, C** and **D**. Choose the correct answer. Record your choice in **soft pencil** on the separate answer sheet.

Read very carefully the instructions on the answer sheet.

INFORMATION FOR CANDIDATES

There are **forty** questions in this paper. Each correct answer will score **one** mark. Any rough working should be done on this question paper.

This question paper consists of 22 printed pages.

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DATA

speed of light in free space	$c = 3.00 \times 10^8 \text{ ms}^{-1}$
permeability of free space	$\mu_0 = 4\pi \times 10^{-7} \text{ Hm}^{-1}$
permittivity of free space	$\epsilon_0 = 8.85 \times 10^{-12} \text{ Fm}^{-1}$ ($1/4\pi\epsilon_0 = 8.99 \times 10^9 \text{ mF}^{-1}$)
elementary charge	$e = 1.60 \times 10^{-19} \text{ C}$
the Planck constant	$h = 6.63 \times 10^{-34} \text{ Js}$
unified atomic mass unit	$1 \text{ u} = 1.66 \times 10^{-27} \text{ kg}$
rest mass of electron	$m_e = 9.11 \times 10^{-31} \text{ kg}$
rest mass of proton	$m_p = 1.67 \times 10^{-27} \text{ kg}$
molar gas constant	$R = 8.31 \text{ JK}^{-1} \text{ mol}^{-1}$
the Avogadro constant	$N_A = 6.02 \times 10^{23} \text{ mol}^{-1}$
the Boltzmann constant	$k = 1.38 \times 10^{-23} \text{ JK}^{-1}$
gravitational constant	$G = 6.67 \times 10^{-11} \text{ Nm}^2 \text{ kg}^{-2}$
acceleration of free fall	$g = 9.81 \text{ ms}^{-2}$

FORMULAE

uniformly accelerated motion	$s = ut + \frac{1}{2}at^2$
	$v^2 = u^2 + 2as$
work done on/by a gas	$W = p \Delta V$
gravitational potential	$\phi = -Gm/r$
hydrostatic pressure	$p = \rho gh$
pressure of an ideal gas	$p = \frac{1}{3} \frac{Nm}{V} \langle c^2 \rangle$
simple harmonic motion	$a = -\omega^2 x$
velocity of particle in s.h.m.	$v = v_0 \cos \omega t$
	$v = \pm \omega \sqrt{(x_0^2 - x^2)}$
Attenuation of x-rays	$I = I_0 e^{-\mu x}$
electric potential	$V = \frac{Q}{4\pi\epsilon_0 r}$
capacitors in series	$1/C = 1/C_1 + 1/C_2 + \dots$
capacitors in parallel	$C = C_1 + C_2 + \dots$
energy of charged capacitor	$W = \frac{1}{2} QV$
electric current	$I = Anvq$
resistors in series	$R = R_1 + R_2 + \dots$
resistors in parallel	$1/R = 1/R_1 + 1/R_2 + \dots$
Hall voltage	$V_H = \frac{BI}{ntq}$
alternating current/voltage	$x = x_0 \sin \omega t$
radioactive decay	$x = x_0 \exp(-\lambda t)$
decay constant	$\lambda = \frac{0.693}{t_{\frac{1}{2}}}$

1. Which equation relating momentum, p , kinetic energy, E , and mass, m is homogeneous?
- A $p^2 = 2Em$
- B $p = 2Em$
- C $p^2 = \frac{2E}{m}$
- D $p = \frac{2E}{m^2}$
2. What is the cause of the scatter of points about the line of best fit on a graph?
- A zero error
- B parallax error
- C random error
- D systematic error
3. The zero error on the micrometer is $+0.04$ mm. The reading is 3.61 mm when the micrometer is used to measure the diameter of a rod.
- What is the correct diameter of the rod?
- A 3.57 mm
- B 3.59 mm
- C 3.63 mm
- D 3.65 mm



4. A car slows down uniformly from 30 ms^{-1} to 20 ms^{-1} while travelling a distance of 50 m along a track.

What is the deceleration?

- A -5.0 ms^{-2}
 B 5.0 ms^{-2}
 C 10.0 ms^{-2}
 D -10.0 ms^{-2}

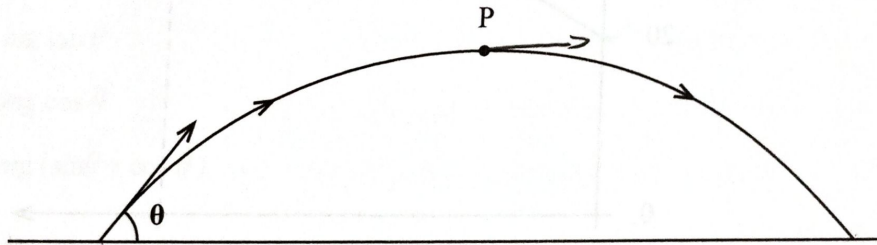
$$v^2 = u^2 + 2as$$

$$\frac{20^2 - 30^2}{2(50)} = a$$

$$acc = -5$$

$$= 5 \text{ ms}^{-2}$$

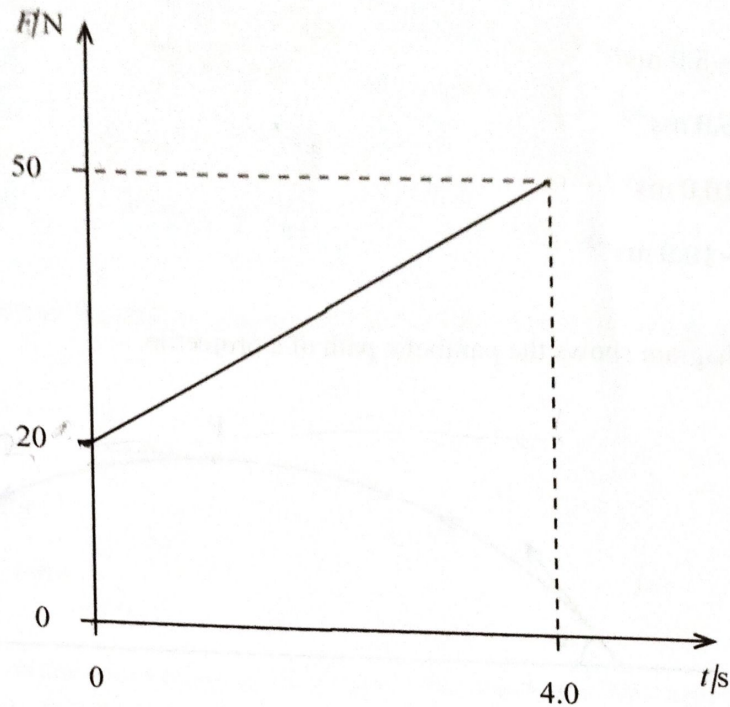
5. The diagram shows the parabolic path of a projectile.



Which one is true about its motion at the highest point, P?

- A velocity is constant
 B acceleration is zero
 C horizontal velocity is zero ✗
 D vertical velocity is zero ✓

6. An object with initial momentum, P , is subjected to a force, F , which varies with time, t , as shown in the diagram.



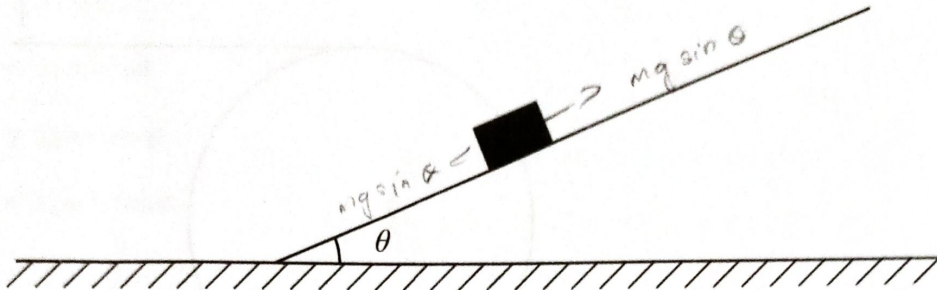
What is the final momentum of the object after 4.0 s, given that F is in the same direction as P ?

- A $(P - 280) \text{ kgms}^{-1}$
- B $(P - 140) \text{ kgms}^{-1}$
- C $(P + 140) \text{ kgms}^{-1}$
- D $(P + 280) \text{ kgms}^{-1}$

$$\begin{aligned}
 p_i &= F \cdot t \\
 &= (50 - 20) \times 4 \\
 &= 80
 \end{aligned}$$



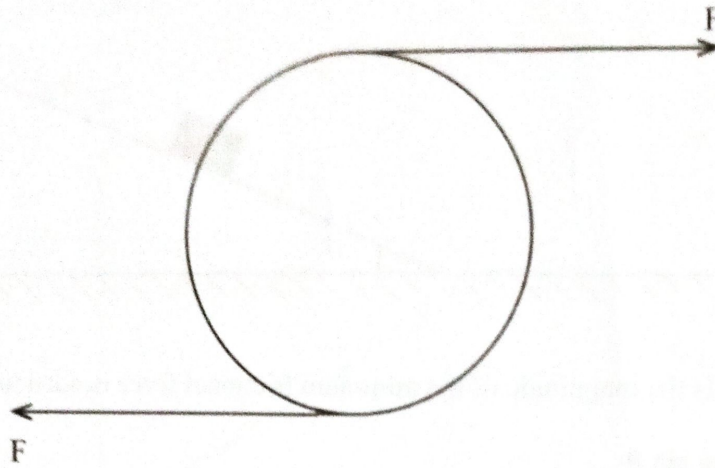
7. The diagram shows a body of mass, m , at rest on a plane inclined at an angle, θ , to the horizontal.



What is the magnitude of the minimum frictional force needed to keep the body at rest?

- A $mg \sin \theta$
 - B $mg \tan \theta$
 - C $mg \cos \theta$
 - D $mg (\sin \theta + \cos \theta)$
8. In uniform circular motion, which physical quantity remains constant?
- A angular velocity
 - B linear velocity
 - C angular displacement
 - D centripetal acceleration

9. The diagram shows two equal and opposite forces acting on an object.



What is the effect of the forces on the object?

- A anticlockwise rotation and translation to the left
 - B clockwise rotation and translation to the left
 - C anticlockwise rotation
 - D clockwise rotation
10. A cellphone battery is being recharged from a solar panel.

What is the overall energy conversion?

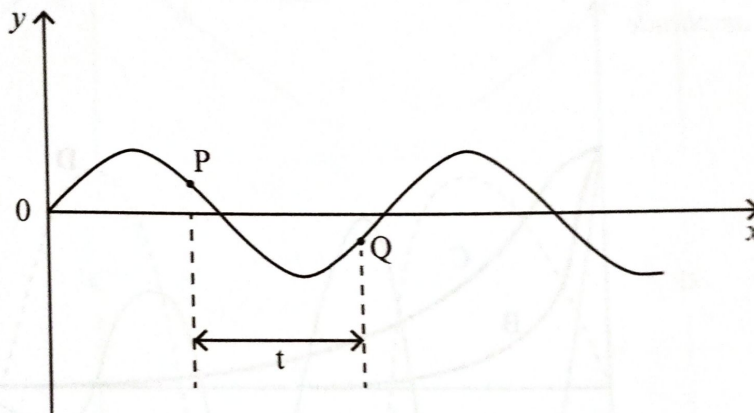
- A heat $\rightarrow$ chemical
- B light $\rightarrow$ electrical
- C light $\rightarrow$ chemical
- D heat $\rightarrow$ electrical



11. Which one is a solution to the equation: $a = -\omega^2 x$?

A $x = x_0 \sin \omega t$
 B $x = x_0 \sin^2 \omega t$
 C $x = x_0 \omega \sin \omega t$
 D $x = x_0 \omega^2 \sin \omega t$

12. The graph shows the variation of displacement, y , with distance, x , for a wave of wavelength, λ .



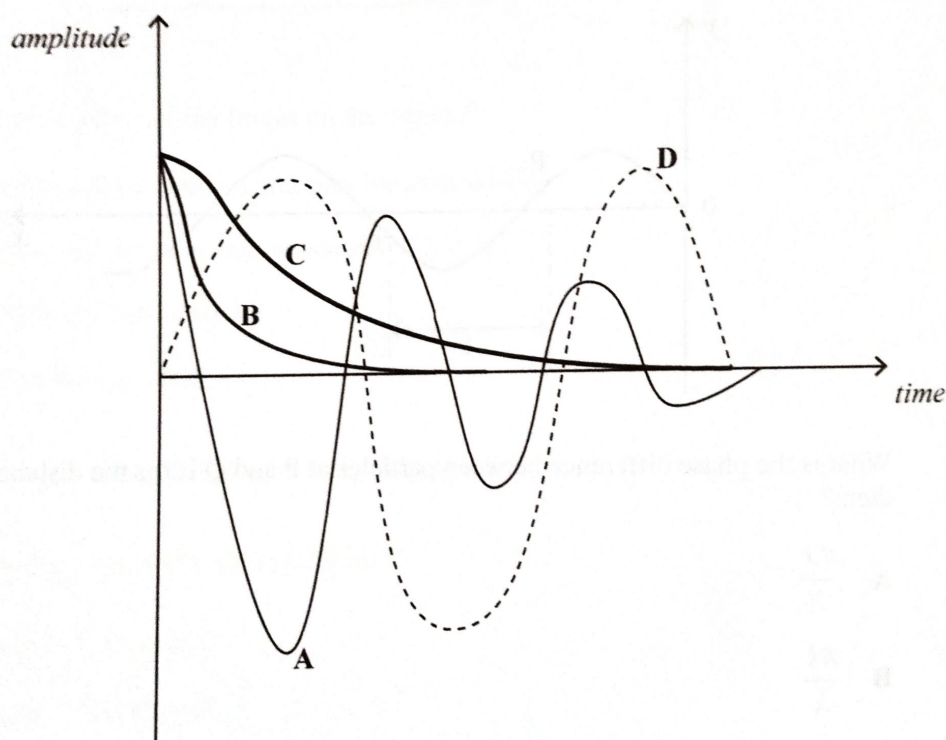
What is the phase difference between particles at P and Q if t is the distance between them?

A $\frac{\pi x}{\lambda}$
 B $\frac{\pi t}{\lambda}$
 C $\frac{2\pi x}{\lambda}$
 D $\frac{2\pi t}{\lambda}$

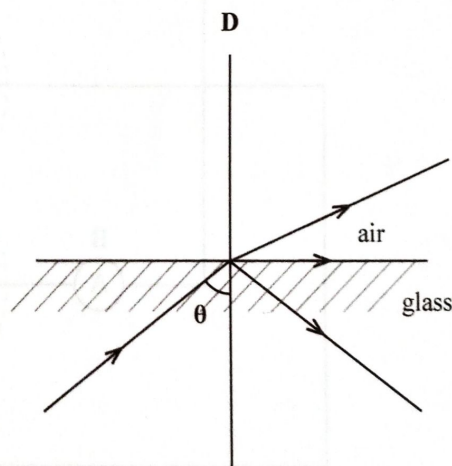
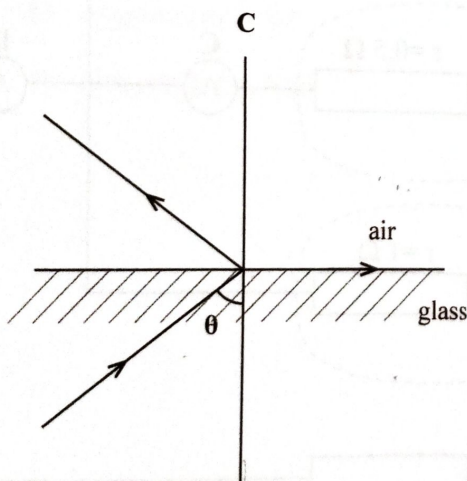
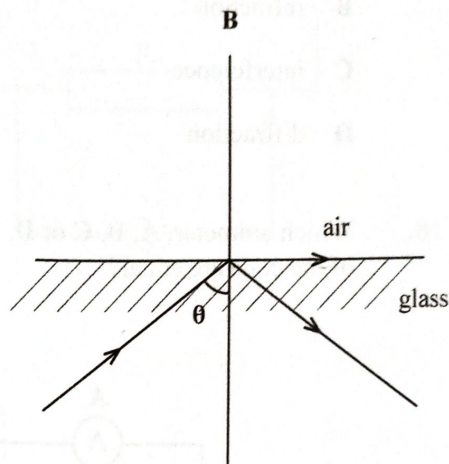
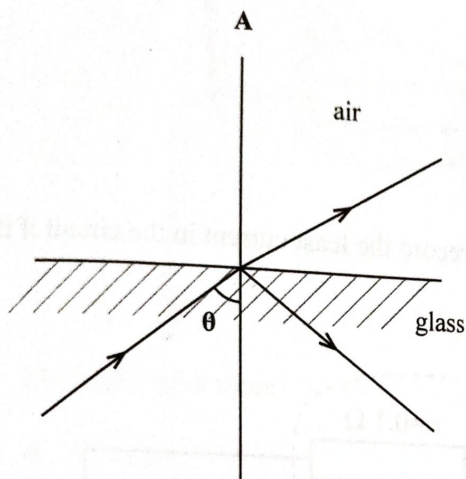
13. Which one is an application of microwaves?

- A sterilization
- B radiotherapy
- C satellite communication
- D imaging of internal structures

14. Which graph, A, B, C or D, shows damping in moving coil meters?



15. A light ray is travelling from glass to air. The angle of incidence, θ , is increased to the critical value.
Which ray diagram, A, B, C or D, correctly shows this situation?



16. Which statement is true about Young's double slit experiment?

- A When the slit separation is large, the fringe separation is small.
B When the slits are narrow, the fringe separation is large.
C When the distance between the slits and the screen is large, the fringe separation is small.
D When the distance between the slits and the screen is small, the fringe separation is large.

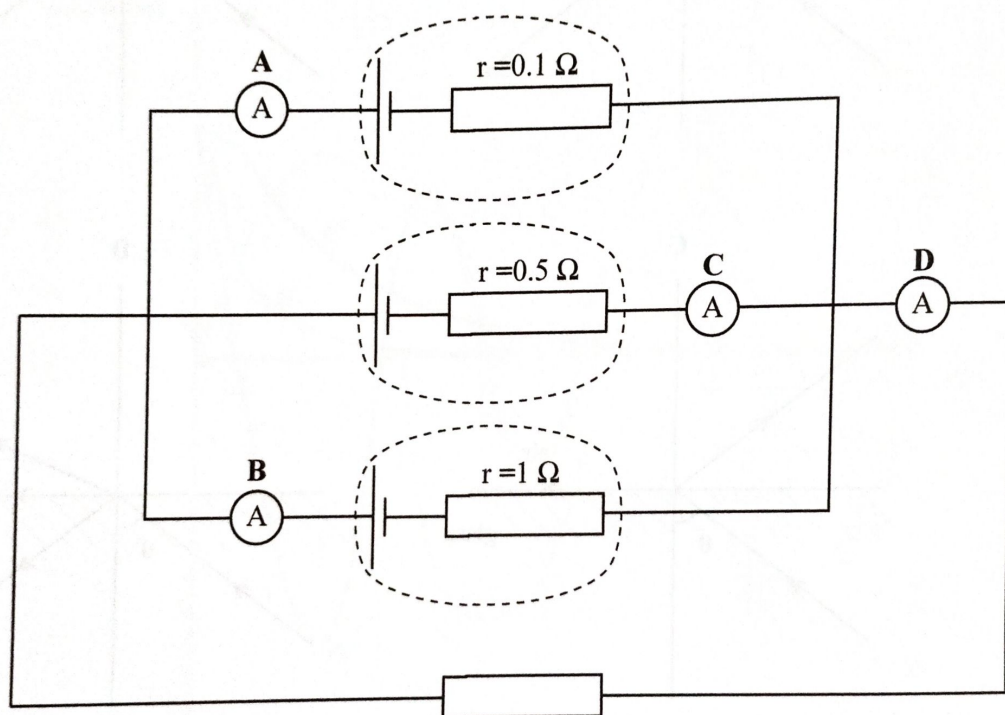
$$y = \frac{\lambda D}{d}$$



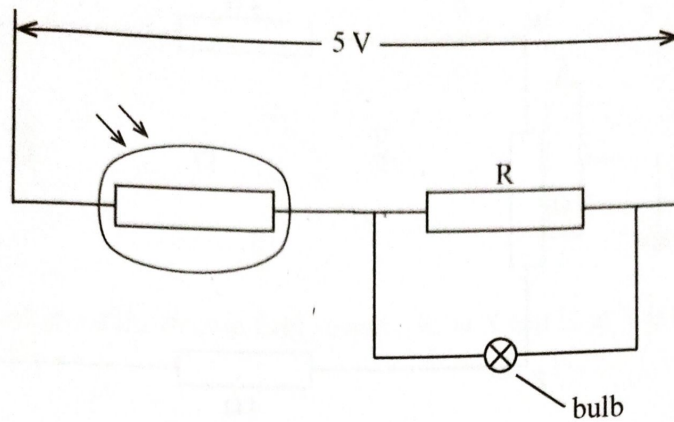
17. Which one is a necessary condition for the formation of a stationary wave?

- A dispersion
- B refraction
- C interference
- D diffraction

18. Which ammeter, A, B, C or D, would record the least current in the circuit if the emfs of the cells are equal?



19. The diagram shows a sensor in a circuit.

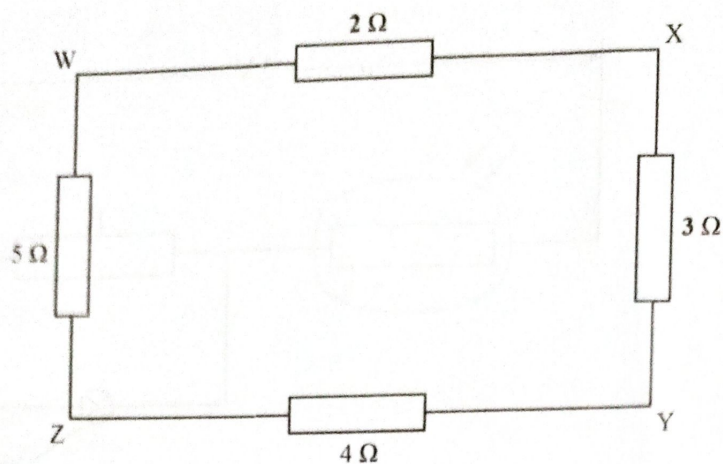


The bulb lights when

- A it is dark.
- B there is light.
- C temperature is low.
- D temperature is high.



20. The diagram shows an arrangement of resistors.



Between which two points is the effective resistance maximum?

- A W and X
B W and Y
C X and Z
D W and Z

$$R_{WZ} = \left(\frac{1}{3} + \frac{1}{2} \right)^{-1} + \left(\frac{1}{5} + \frac{1}{4} \right)^{-1}$$

21. What happens to the force on the test charge when the distance is doubled, if a test charge is positioned at a distance, r , from a point charge?

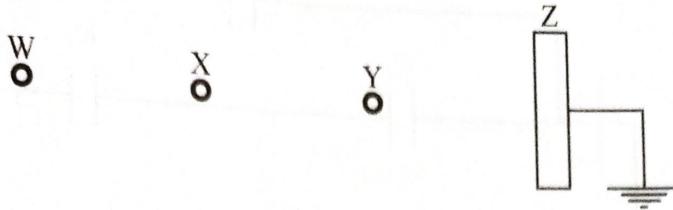
- A halved
B doubled $\times$
C increased by a factor of four $\times$
D reduced by a factor of four

$$E = \frac{F}{q}$$

$$F \propto \frac{Qq}{r^2}$$



22. An earthed metal plate, Z is placed in front of a charged particle at W. Points X and Y are between W and Z.

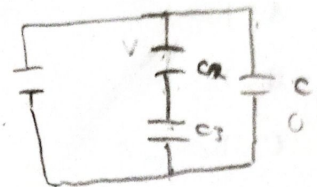
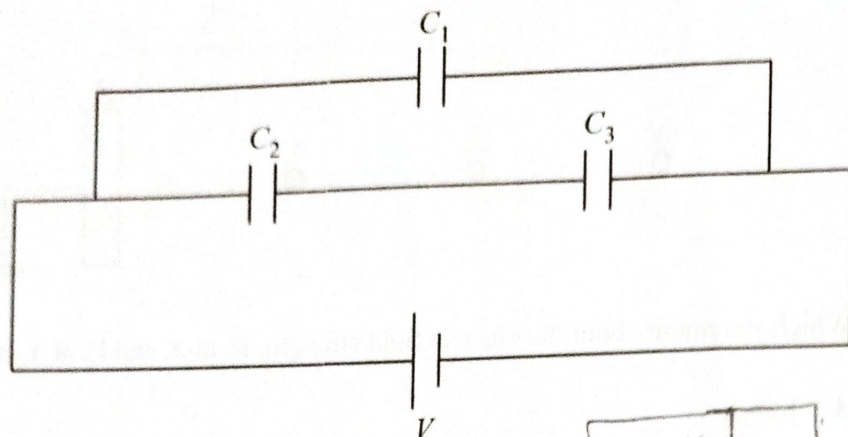


Which statement about the electric field strength, E_x at X and E_y at Y is correct?

- A $E_x = 0$
- B $E_y = 0$
- C $E_x > E_y$
- D $E_x < E_y$
23. In a vacuum, a charged particle experiences a force only while it is in motion. What is the likely cause of this force?
- A air resistance
- B magnetic field $F = Bev$
- C electric field $F =$
- D gravitational field
24. What is the diameter of the circular path of an electron moving with a speed of $1.5 \times 10^5 \text{ ms}^{-1}$ in a magnetic field of flux density 6.0 mT ?
- A $1.42 \times 10^{-4} \text{ m}$
- B $2.84 \times 10^{-4} \text{ m}$
- C $7.03 \times 10^3 \text{ m}$
- D $1.41 \times 10^4 \text{ m}$



25. The diagram shows an arrangement of identical capacitors.



Which statement about the capacitors is correct?

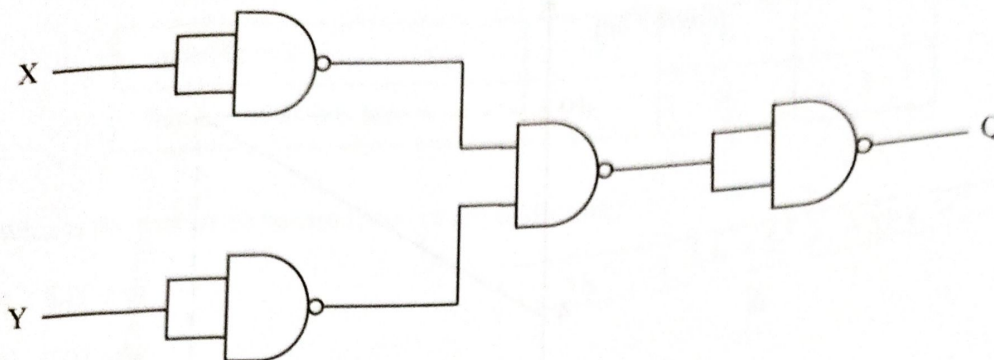
- A C_1 and C_2 store the same amount of charge
 - B C_1 and C_3 store the same amount of energy
 - C C_2 and C_3 store the same amount of energy
 - D C_1 , C_2 and C_3 store the same amount of charge
26. A smoothing capacitor of capacitance, C , and a resistor of resistance, R , are connected in parallel at the output of a rectifier.

What adjustment to C and R can be made to ensure a smoother output across the resistor?

	C	R
A	reduce	increase
B	increase	increase
C	increase	reduce
D	reduce	reduce



27. The diagram shows an arrangement of NAND gates only.



Which single logic gate is represented by the circuit?

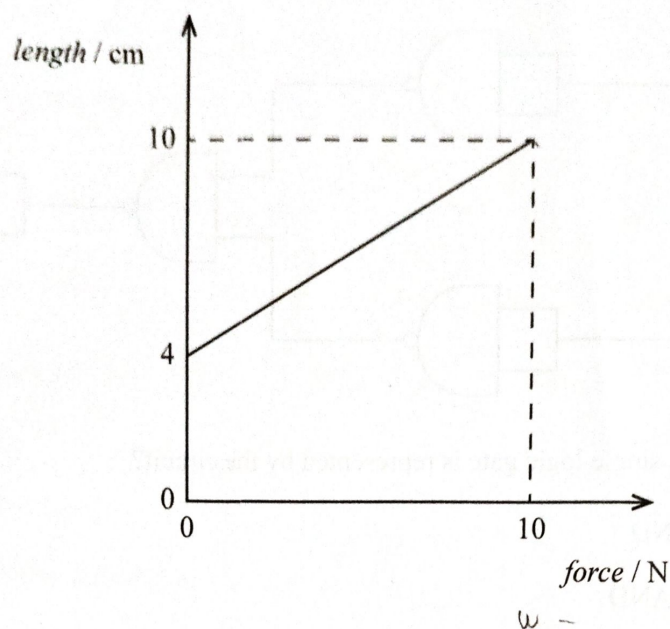
- A AND
 - B NAND
 - C OR
 - D NOR
28. Which statement correctly describes the latent heat of vaporisation?
- A heat energy required to change the state of a substance from liquid to gas per unit mass
 - B heat energy required per unit mass to change the state of a substance from liquid to gas at a constant temperature
 - C heat energy absorbed per unit mass to change the state of a substance from gas to liquid at a constant temperature
 - D heat energy absorbed to change the state of a substance from liquid to gas at a constant temperature

$$Q = m l_v$$

$$l_v = \frac{Q}{m}$$



29. The graph shows the variation of *length* of a spring with *force* exerted on it.



What is the work done in stretching the spring?

- A 0.30 J
 - B 0.50 J
 - C 0.60 J
 - D 0.70 J
30. Which one is not an assumption of the kinetic theory of gases?
- A All molecules in a container move randomly at the same speed.
 - B The collision of molecules with the walls of the container is perfectly elastic.
 - C The duration of intermolecular collisions is negligible.
 - D There are no intermolecular forces.



31. The speeds of molecules are distributed as shown in the table.

speed in m/s	15	6	7
number of molecules	1	4	3

What is the root mean square speed of the molecules?

- A 6.22 m/s
B 8.03 m/s
C 38.8 m/s
D 64.5 m/s

$$\sqrt{\frac{15^2 + (4 \times 6^2) + (3 \times 7^2)}{8}}$$

32. Water in a pipe is initially at rest, exerting a pressure of 5.0×10^5 Pa. When the water begins to flow, the pressure decreases to 4.5×10^5 Pa. Assuming the density of water is 1000 kg/m^3 and the height of the pipe remains constant, what is the velocity of the water?

- A 22.4 m/s
B 14.1 m/s
C 10.0 m/s
D 7.1 m/s

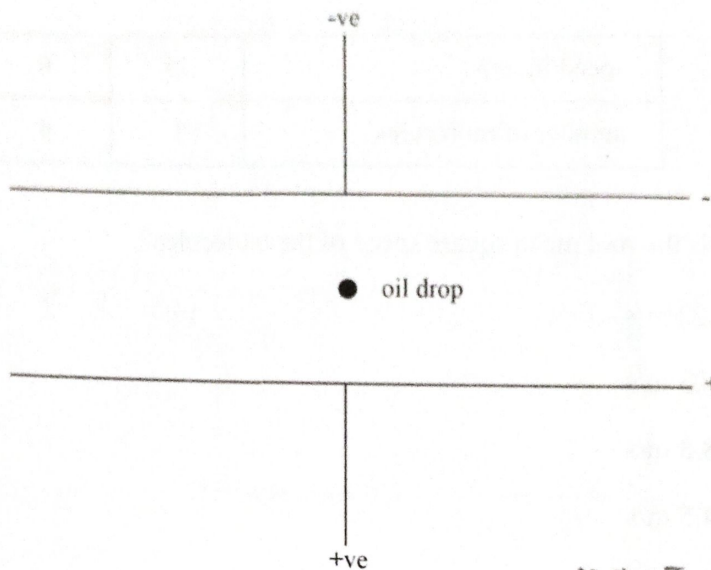
$$P - P = \frac{1}{2} \rho v^2$$

$$v = \sqrt{\frac{2(P - P)}{\rho}}$$

$$= \sqrt{2(\quad)}$$



33. The diagram shows a positively charged oil drop held stationary between horizontal charged metal plates.



The positive charge on the oil drop is reduced slightly.

$$mg = q \frac{V}{d}$$

What change should be made to keep the oil drop stationary?

- A increase the potential difference between the plates ✓
 - B increase separation of the plates ✗
 - C reverse the charges on the plates ✗
 - D allow the drop to move closer to the negative plate ✗
34. In Rutherford's alpha scattering experiment, the large angle deviation of alpha particles can be explained by
- A scattering from free electrons.
 - B diffraction effects due to the crystal lattice.
 - C random reflections from the surface of the metal foil.
 - D scattering from small, dense and positively charged regions within the atom.



35. Which statement is true about the photoelectric effect?
- A The intensity of the radiation is directly proportional to the kinetic energy of the emitted electrons. $\times$
 - B The number of electrons emitted per second depends on the frequency of the incident radiation. $\times$
 - C Emission commences instantly when the surface is irradiated with radiation above threshold frequency. $\checkmark$
 - D All emitted electrons have the same kinetic energy. $\times$

36. A 60 g sample of an iodine radioisotope has a half-life of 8 days.
What mass of the sample will have decayed after 24 hours?

- A 55.02 g
- B 52.50 g
- C 7.50 g
- D 4.98 g

$$m = 60 e^{-0.8664 \times 1}$$

$$\lambda = \frac{\ln 2}{8}$$

$$= 0.08664/\text{day}$$

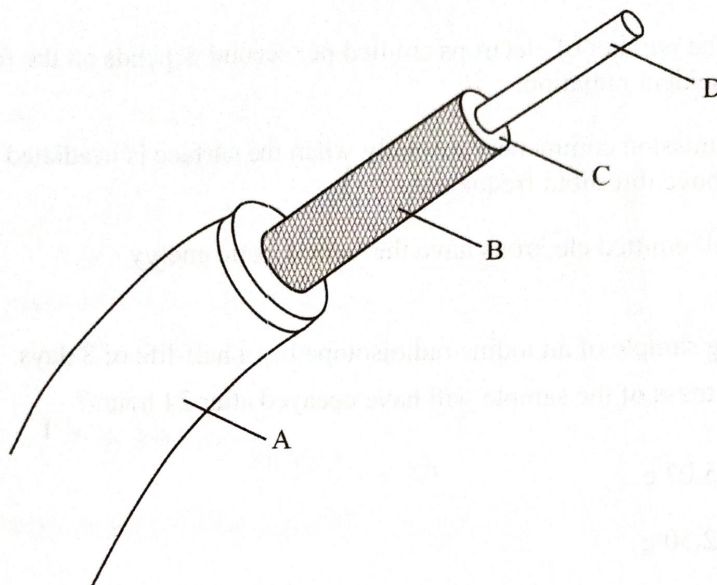
37. Which type of radiation is most ionising?

- A X rays
- B gamma rays
- C ultra violet rays
- D beta particles

$$2mivux9$$



38. The diagram shows a coaxial cable.
Which part shield the inner core from electromagnetic noise?



39. Which one is **NOT** an advantage of optic fibres over coaxial cables?

- A less weight
- B higher security
- C lower bandwidth
- D less interference

40. The gain of an amplifier is 30 dB and the output power is 80 mW.
What is the input power?

- A 80 mW
- B 8 mW
- C 0.8 mW
- D 0.08 mW

$$30 = 10 \log \left(\frac{80 \times 10^{-3}}{P_{in}} \right)$$

$$10^{3/0} = 80 \times 10^{-3}$$

$$P_{in} = \frac{80 \times 10^{-3}}{10^3}$$

$$0,00008$$



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